# Examine why advances in science are often the result of collaboration or build on the work of others

## Learning intention

- examine why advances in science are often the result of collaboration or build on the work of others.

## Teach and model

- researching why European naturalists and scientists first thought the platypus was a faked animal.
- exploring how understanding of light and optics has developed by comparing the ideas of Plato, Euclid, Ptolemy, Ibn al-Haytham and Roger Bacon.
- researching how the recent discovery of a biofluorescent flying squirrel led to discoveries of more fluorescent mammals, such as wombats, bilbies.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.